Implementation of Verse to Simulate and Validate Satellite Dynamics.

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Introduction

- After launch vehicle separation, satellites enter a uncontrolled tumble state which can prevent communication to ground station due to high angular velocity.
- To stabillize the satellite, reaction wheels and magnetorquers and used in control algorithms to decrease angular velocity during the orbit, and reach the desired attitude.
- This simulation highlights a hybrid mode satellite with a detumble phase and a pointing phase.

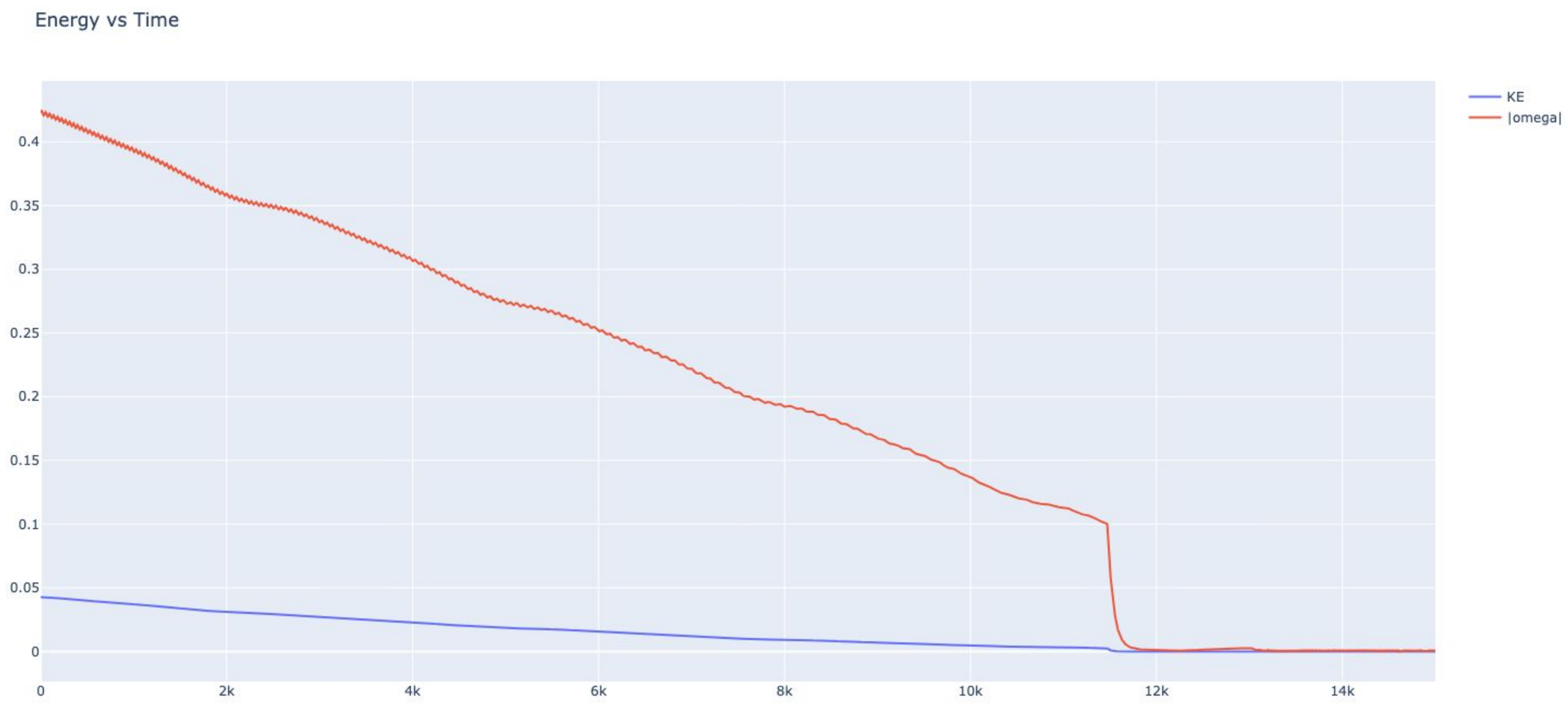
Environmental Modeling

- Four external torques acting on the satellite body are simulated:
  - Gravity-Gradient Torque: Non-uniform force of gravity acts on the satellite body.
  - Magnetic Torque: Residual dipole moment interacting with Earth's magnetic field.
  - Aerodynamic Drag Torque: Atmospheric Drag at LEO around the lever arm between CoP and CoG.
  - Solar Radiation Pressure Torque: Photon pressure on the sun exposed face.

Hybrid Modes

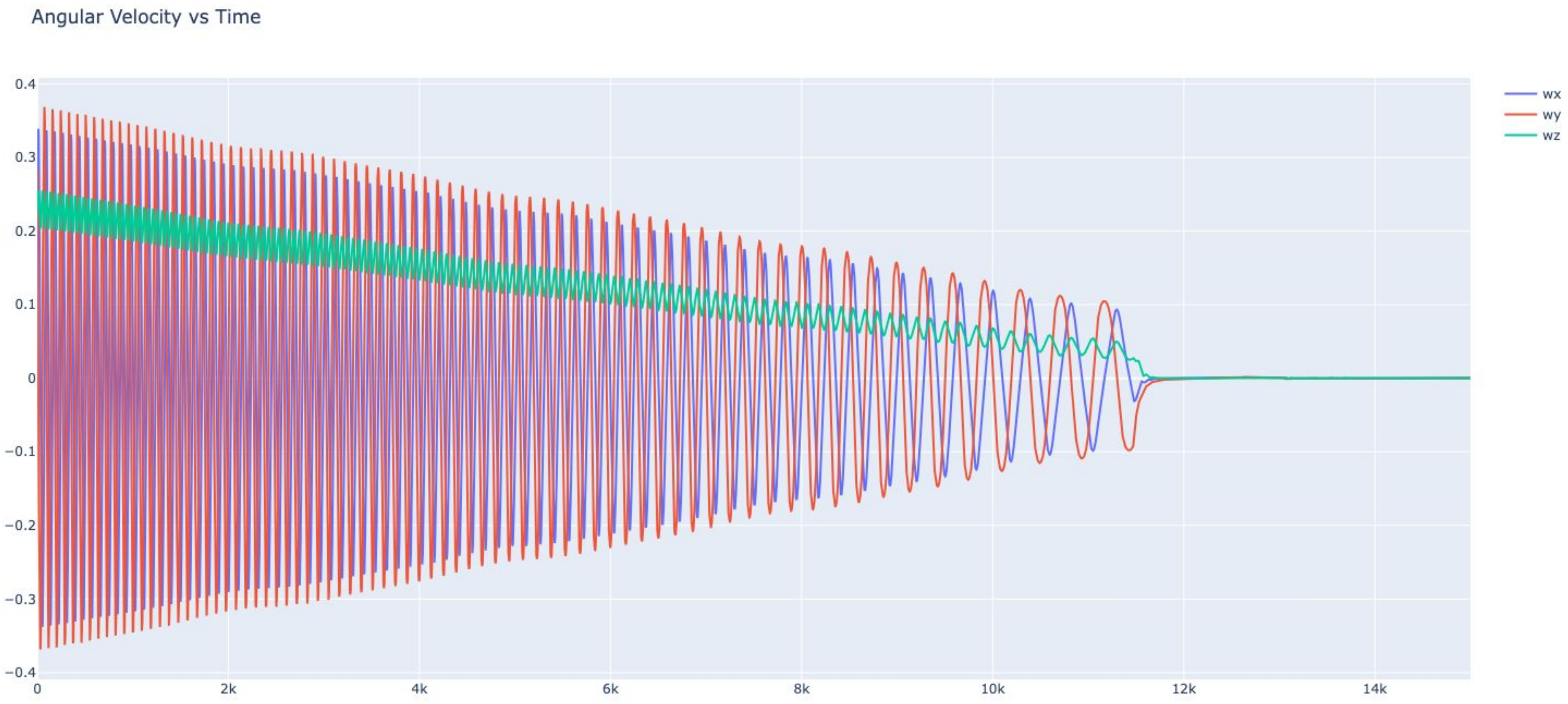
- B-dot Control Algorithm
  - Creates a magnetic dipole opposite to the rate of change of the magnetic field to reduce angular velocities.
  - Control input acts perpendicular to magnetic field, resulting in a decaying oscillation behavior.
- PID Pointing Control Algorithm
  - Actuates Reaction Wheels to reach a desired quaternion state.
  - Gain values determined by a paper, using satellites inertia and integration step.
- Change in mode based on the magnitude of angular velocity.
  - w < 0.1 rad/s → Pointing
  - w > 0.5 rad/s → Detumble

Simulation Results - Energy vs Time



- Magnitude of angular velocity steadily drops to 0 during the simulation. The change is slope at ~ 11.8k is when the Satellite goes from B-dot to PID control.
- No energy is injected into the system (Blue).

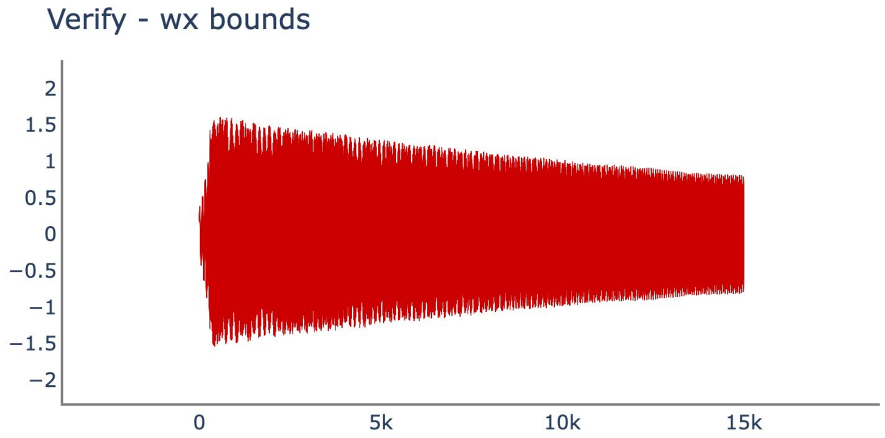
Simulation Results - Omega vs Time



- Oscillating during the Detumble phase, steadily decreasing with B-dot Control.
- At ~11.8k, the angular velocities are low enough where PID control takes over and brings them to 0.

Verification Results - Reachability of Angular Velocity

- Simulates an offset of initial angular velocity values to verify controllability in multiple scenarios.



- No matter the initial conditions, Verse is showing that the controller is working, and the angular velocity is bounded and converging.
- This proves a successful verification of the dynamics and controller.